Individual Progress Report: Week 3

Group Topic: Urban Air Quality & Environmental Health **Lab No.** Week 3

Git Repo/ Colab Link: https://github.com/Cristy16/BDA---Final-Project **Date:** 05/08/26

I. Objectives

General Objectives

My main goal for this phase was to build the technical foundation for our machine learning pipeline. This involved setting up the environment, justifying our choice of model, and testing how well the Random Forest Regressor actually performs on our cleaned dataset.

Individual Objective (Setup and Justification)

- **Environment & Data Prep:** I needed to set up a Jupyter Notebook to handle the Phase 5 data, specifically organizing the features (X) and the target variable (y) for predicting PM2.5 levels.
- **Methodology:** I aimed to provide a solid, data-backed reason for why we are using Regression instead of something like Clustering or Classification.
- **Validation:** My goal was to test the model's accuracy using a large test set (over 134,000 rows) and create visuals that make the results easy to understand.

II. Tasks Completed

- **Built the Notebook:** I put together a standalone Jupyter Notebook that takes the Phase 5 CSV and organizes the inputs, specifically CO, NO_2, SO_2, and location, to predict PM2.5.
- **Defined the Strategy:** I wrote a section explaining why Regression is the best fit for this project. While other methods just group data or label it "good" or "bad," Regression lets us see the actual numerical relationship between different gases and pollution levels.
- **Model Tuning & Metrics:** I ran the Random Forest Regressor with specific settings (n_estimators=50, min_samples_split=10). To see how it did, I calculated the R^2, RMSE, and MAE against our test data.
- **Visual Documentation:** I created a scatter plot to compare actual vs. predicted values and a chart showing which features mattered most to the model. I also made sure to include notes on data ethics and what our metrics actually mean.

III. Tasks to be Completed

Now that I've finished the setup, justification, and initial evaluation, I'm handing over the final notebook so we can merge it into the main Git repository for the team.

IV. Results and Analysis

Model Performance Summary

Metric	Value	Interpretation
R^2 (R-Squared)	0.5237	The model explains about 52% of the variance in PM2.5.
RMSE	7.2692 \mu g/m^3	This is the average deviation from the actual values.
MAE	5.3445 \mu g/m^3	The average absolute error across the whole test set.

Analytical Insights

- **Why Regression Works:** Using Regression was definitely the right call. Because CO and NO_2 are so closely linked (0.78 correlation), the model can pick up on subtle patterns that a simple Classification model would have missed.
- **What Drives the Predictions:** Carbon Monoxide (CO) turned out to be the most important factor for the model. This makes sense because it matches what we found during our earlier data exploration, showing the model is learning real chemical relationships.
- **Where it Struggles:** The model is really accurate when pollution is low or moderate, but the scatter plot shows it struggles more during extreme spikes. I've noted that this is likely because we don't have weather or traffic data in our current set to explain those sudden jumps.

V. Challenges and Solutions

While working on the evaluation code, I kept hitting a *NameError* for *y_test* and other variables. I realized it was because I was trying to run specific cells without the ones before them being loaded. I fixed this by setting up a "Run All" protocol to make sure the environment, data splits, and parameters are all initialized in the right order before the evaluation starts.

VI. Conclusion

The move to Phase 6 went smoothly. By setting up the environment properly and proving why Regression is our best tool, the team now has a solid foundation for forecasting air quality. The model is performing much better than our baseline, and the notebook is ready to be integrated into the group repo.